

This is for a cartesian language. I'm not sure how quadraticity plays into the unification algorithm.

First-order terms consist of variables and constants applied to lists of terms. A list of terms is either empty ($\cdot$) or a cons.

$$\frac{x \in \Gamma}{\Gamma \vdash x} \quad \frac{\Gamma \vdash \vec{M}}{\Gamma \vdash F(\vec{M})} \quad \frac{}{\Gamma \vdash \cdot} \quad \frac{\Gamma \vdash M \quad \Gamma \vdash \vec{M}}{\Gamma \vdash M, \vec{M}}$$

A context is $\cdot$ or Γ, x . A substitution is

$$\frac{}{\Gamma \vdash \cdot : \cdot} \quad \frac{\Gamma \vdash \theta : \Gamma' \quad \Gamma \vdash M}{\Gamma \vdash \theta, M/x : \Gamma', x}$$

Identity and composition are admissible:

$$\frac{}{\Gamma \vdash 1_\Gamma : \Gamma} \quad \frac{\Gamma \vdash \theta : \Gamma' \quad \Gamma' \vdash \theta'' : \Gamma''}{\Gamma \vdash \theta''[\theta] : \Gamma''}$$

The unification judgements have the form

- $M \doteq N \mid \theta$ where $\Gamma \vdash M$ and $\Gamma \vdash N$ and $\Gamma' \vdash \theta : \Gamma$. Operationally, we think of M and N and Γ as inputs and Γ' and θ as outputs, so unification produces a new context and a substitution for the original variables from it.
- $\vec{M} \doteq \vec{N} \mid \theta$ where $\Gamma \vdash \vec{M}$ and $\Gamma \vdash \vec{N}$ and $\Gamma' \vdash \theta : \Gamma$.

Here are the rules (I put in some type annotations that might be helpful):

$$\frac{}{\cdot \doteq \cdot \mid 1_\Gamma} \quad \frac{M \doteq N \mid (\Gamma' \vdash \theta : \Gamma) \quad \vec{M}[\theta] \doteq \vec{N}[\theta] \mid (\Gamma'' \vdash \theta' : \Gamma')}{M, \vec{M} \doteq N, \vec{N} \mid \theta[\theta']}$$

$$\frac{\Gamma \vdash M}{x \doteq M \mid (\Gamma \vdash (1_\Gamma, M/x) : \Gamma, x)} \quad \frac{\vec{M} \doteq \vec{N} \mid \theta}{F(\vec{M}) \doteq F(\vec{N}) \mid \theta}$$